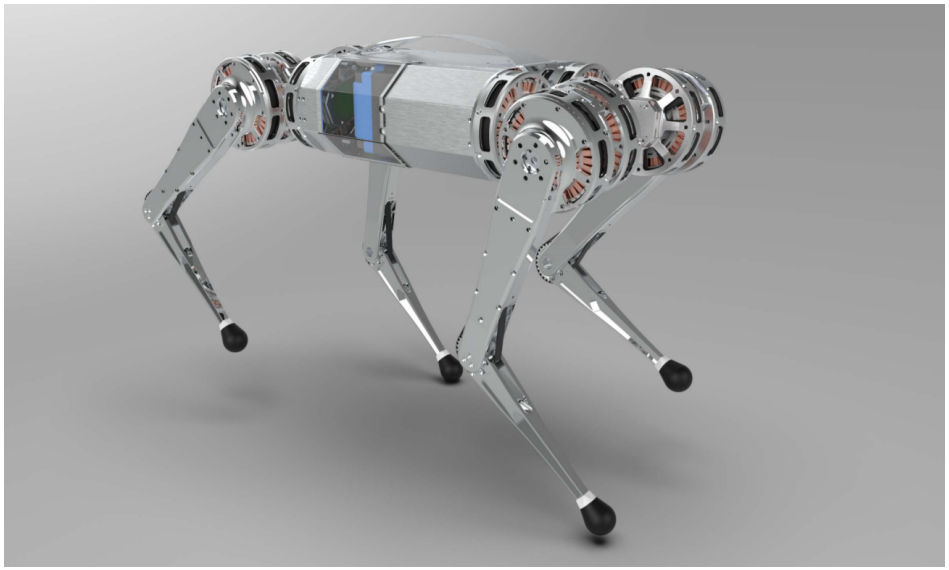
Quadruped Robots

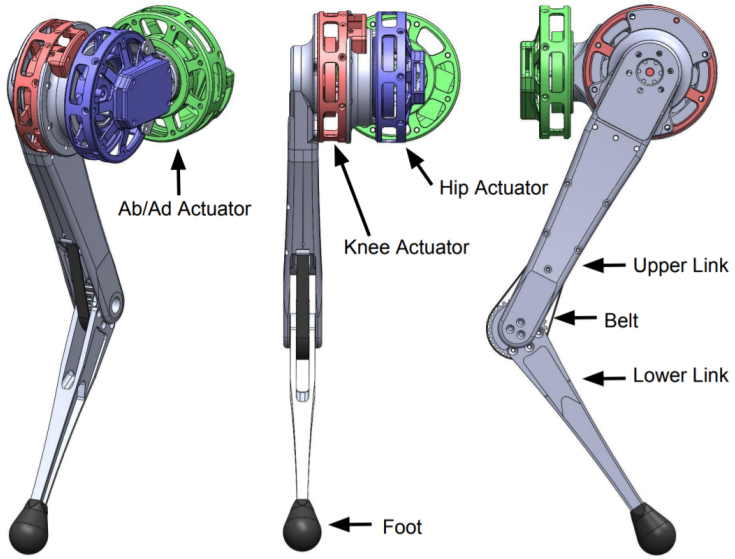
Xu Yang

December 1, 2021

MIT Cheetah 3

The Cheetah-Software repository can be found here.





- Motors (including drives and encoders).
- IMU.

MIT Cheetah 3

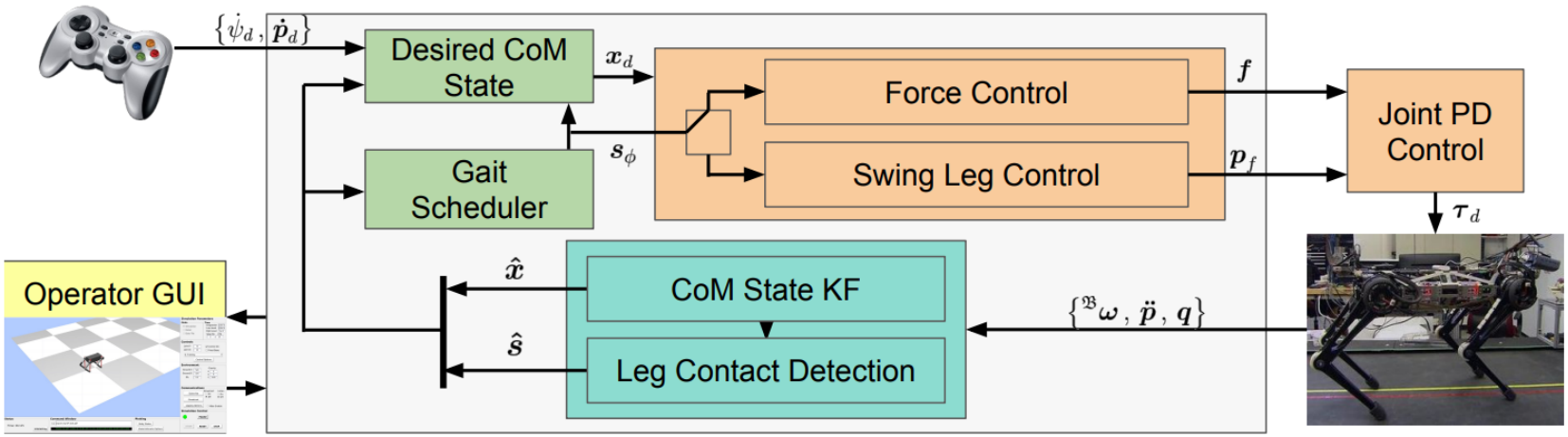


Figure: System Architecture Block Diagram.

- Planning: schedule independent phases of each leg.
- Control: swing leg control and contact force control.
- State estimation.

MIT Cheetah3 - Gait Scheduler

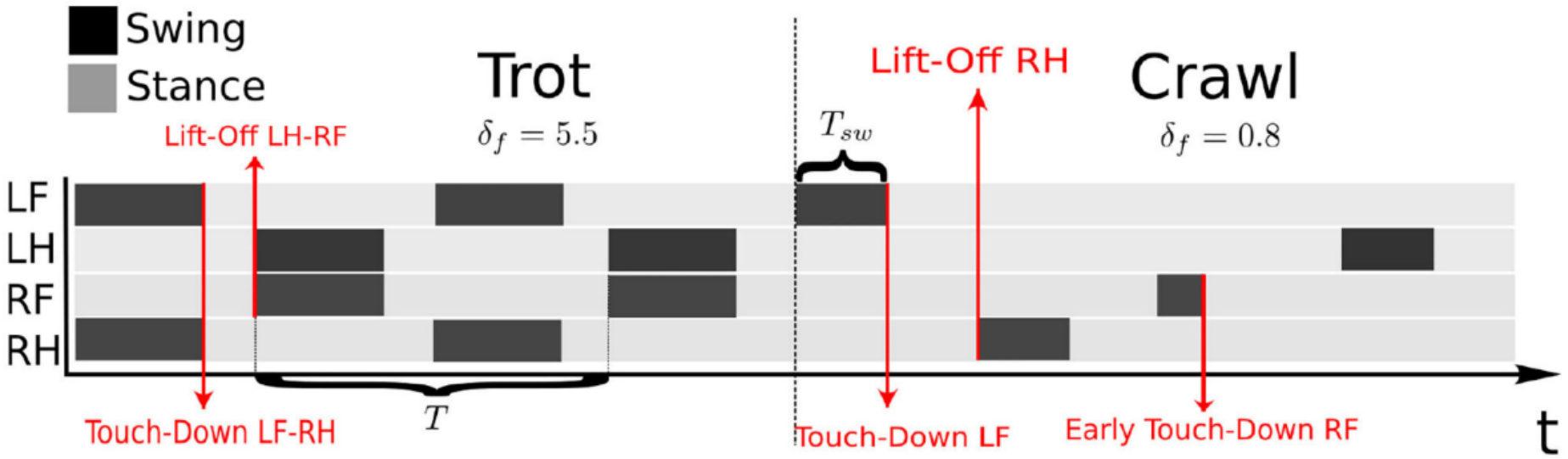


Figure: Schedules of Trot Gait and Crawl Gait

- The footstep gait pattern is nominally specified as a set of time-based phase.

Force Control - Balance Controller

- The key lies in computing expected ground reaction force.

$$\tau_i = -J_i^T F_i.$$

- The leg links have a mass of 2.7kg for all four legs, which is only 6% of the mass of the robot.

$$\underbrace{\begin{bmatrix} I_3 & \cdots & I_3 \\ [p_1 - p_c]\times & \cdots & [p_4 - p_c]\times \end{bmatrix}}_{A} F = \underbrace{\begin{bmatrix} m(\ddot{p}_c + g) \\ I_G \dot{\omega}_b \end{bmatrix}}_{b}.$$

Force Control - Balance Controller

- Desired acceleration and angular acceleration of CoM:

$$\begin{bmatrix} \ddot{p}_{c,d} \\ \dot{\omega}_{b,d} \end{bmatrix} = \begin{bmatrix} K_{p,p}(p_{c,d} - p_c) + K_{d,p}(\dot{p}_{c,d} - \dot{p}_c) \\ K_{p,\omega} \log(R_d R^T) + K_{d,\omega}(\omega_{b,d} - \omega) \end{bmatrix}.$$

- Desired state of CoM:

$$b_d = \begin{bmatrix} m(\ddot{p}_{c,d} + g) \\ I_G \dot{\omega}_{b,d} \end{bmatrix}.$$

- Solve the following QP:

$$\begin{aligned} F^* = &\text{argmin} \quad (AF - b_d)^T S(AF - b_d) + \alpha ||F||^2 + \beta ||F - F^*_{prev}||^2 \\ &s.t. \quad CF \leq d \end{aligned}$$

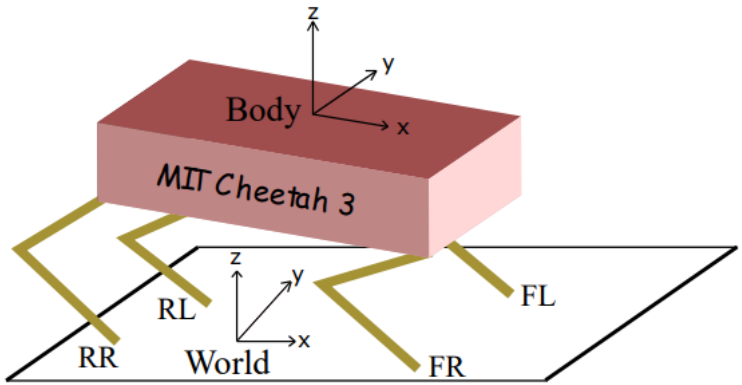
Force Control - MPC

- Suppose the robot's orientation is expressed as a vector of $Z-Y-X$ Euler angles, and $\Theta = [\phi\ \theta\ \psi]^T$, where ψ is the yaw, θ is the pitch and ϕ is the roll. Then the transfrom from body to world coordinates is:

$$R = R_z(\psi)R_y(\theta)R_x(\phi)$$

- For small values of roll and pitch (ϕ, θ), it can be approximated as

$$\left[\dot{\phi}\ \dot{\theta}\ \dot{\psi}\right]^T \approx R_z(\psi)\omega$$



Force Control - MPC

- When the angular velocity ω is small,

$$\sum_{i=1}^{n} r_i \times F_i = \frac{d}{dt}(I_{inertia}\omega) = I\dot{\omega} + \omega \times (I_{inertia}\omega) \approx I_{inertia}\dot{\omega}.$$

-

$$\frac{d}{dt}\begin{bmatrix}\hat{\Theta}\\ \hat{p}\\ \hat{\omega}\\ \hat{\dot{p}}\end{bmatrix} = \begin{bmatrix}0_3 & 0_3 & R_z(\psi) & 0_3\\ 0_3 & 0_3 & 0_3 & 1_3\\ 0_3 & 0_3 & 0_3 & 0_3\\ 0_3 & 0_3 & 0_3 & 0_3\end{bmatrix}\begin{bmatrix}\hat{\Theta}\\ \hat{p}\\ \hat{\omega}\\ \hat{\dot{p}}\end{bmatrix} + \begin{bmatrix}0_3 & \cdots & 0_3\\ 0_3 & \cdots & 0_3\\ \hat{I}_{inertia}^{-1}[r_1]_\times & \cdots & \hat{I}_{inertia}^{-1}[r_n]_\times\\ 1_3/m & \cdots & 1_3/m\end{bmatrix}\begin{bmatrix}F_1\\ \vdots\\ F_n\end{bmatrix} + \begin{bmatrix}0\\0\\0\\g\end{bmatrix}$$

Force Control - MPC

- Simplified robot dynamics:

$$\dot{x}(t) = A_c(\psi)x(t) + B_c(r_1, \cdots, r_n, \psi)u(t).$$

- MPC problem (updated at 30Hz):

$$\min_{x,u} \sum_{i=0}^{k-1} ||x_{i+1} - x_{i+1,ref}||_{Q_i} + ||u_i||_{R_i}$$

$$\begin{aligned} \text{subject to} \quad & x_{i+1} = A_i x_i + B_i u_i, i = 0, \cdots, k-1 \\ & \underline{c}_i \leq C_i u_i \leq \bar{c}_i, i = 0, \cdots, k-1 \\ & D_i u_i = 0, i = 0, \cdots, k-1 \end{aligned}$$

Force Control - MPC+WBIC

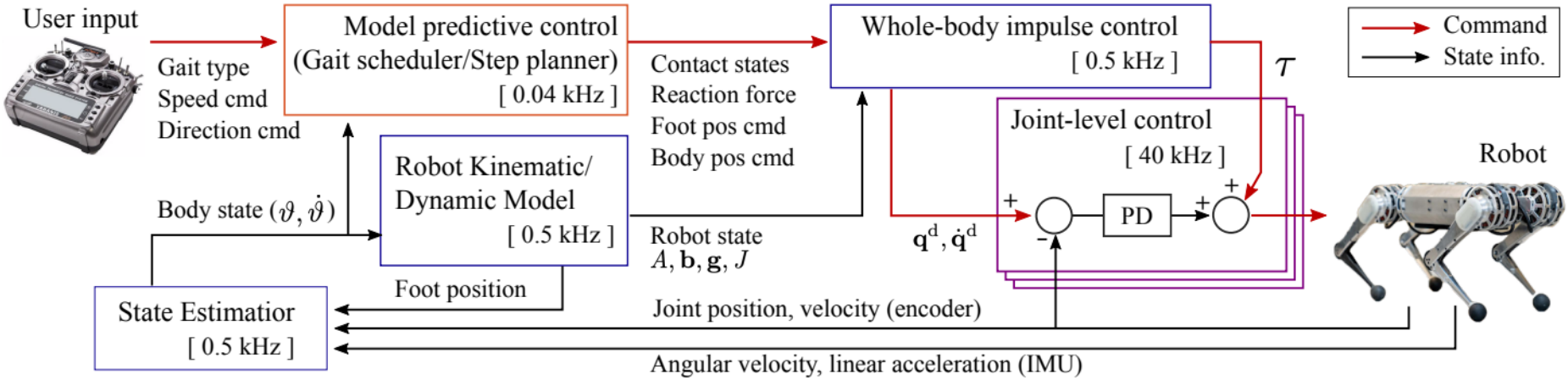


Figure: Overall Control Framework: MPC + Whole-body Impulse Control

Force Control - MPC+WBIC

- QP problem:

$$\min_{\delta_{f_r}, \delta_f} \quad \delta_{f_r}^T Q_1 \delta_{f_r} + \delta_f^T Q_2 \delta_f$$

$$s.t.$$

$$S_f(A\ddot{q} + b + g) = S_f J_c^T f_r \quad \text{(floating base dynamics)}$$

$$\ddot{q} = \ddot{q}^{cmd} + \begin{bmatrix} \delta_f \\ 0_{n_j} \end{bmatrix} \quad \text{(acceleration)}$$

$$f_r = f_r^{MPC} + \delta_{f_r} \quad \text{(reaction forces)}$$

$$Wf_r \geq 0 \quad \text{(contact force constraints)},$$

where $q = \begin{bmatrix} q_f \\ q_j \end{bmatrix}$.

Remark: Multi-task Inverse Differential Kinematics Control

$task_i := \{J_i, w_i^*\}$

- Mapping between joint space velocity $\dot{q}$ and end-effector velocity w_e:

$$w_e = J_{e0}\dot{q}.$$

- Multi-task with equal priority:

$$\dot{q} = \underbrace{\begin{bmatrix} J_1 \\ \vdots \\ J_{n_t} \end{bmatrix}^+}_{\bar{J}} \underbrace{\begin{bmatrix} w_1^* \\ \vdots \\ w_{n_t}^* \end{bmatrix}}_{\bar{w}}.$$

- Multi-task with different priorities: minimization of $||W^{1/2}(\bar{w} - \bar{J}\dot{q})||_2$.

$$\bar{J}^{+W} = (\bar{J}^T W \bar{J})^{-1} \bar{J}^T W.$$

Remark: Multi-task Inverse Differential Kinematics Control

- Taking the Moore-Penrose pseudo inverse:

$$\dot{q} = J_{e0}^{+} w_e^* + N\dot{q}_0, \quad \forall \dot{q}_0,$$

where $N = \mathcal{N}(J_{e0})$ is the null-space projection matrix of J_{e0} and can be given by $N = I - J_{e0}^{+} J_{e0}$.

- For task 1:

$$\dot{q} = J_1^{+} w_1^* + N_1 \dot{q}_0.$$

For task 2:

$$w_2 = J_2 \dot{q} = J_2 (J_1^{+} w_1^* + N_1 \dot{q}_0).$$

Solve for $\dot{q}_0$ and we have

$$\dot{q}_0 = (J_2 N_1)^{+} (w_2^* - J_2 J_1^{+} w_1^*),$$
$$\dot{q} = J_1^{+} w_1^* + N_1 (J_2 N_1)^{+} (w_2^* - J_2 J_1^{+} w_1^*).$$

Swing Leg Control

- The footstep locations are calculated using a linear combination of the Raibert heuristic and a velocity-based feedback term from the capture point formulation.

$$p_{step,i} = p_{h,i} + \underbrace{\frac{T_{c_\phi}}{2}\dot{p}_{c,d}}_{Raibert\ \ Heuristic} + \underbrace{\sqrt{\frac{z_0}{||g||}}(\dot{p}_c - \dot{p}_{c,d})}_{Capture\ \ Point},$$

  where T_{c_ϕ} is the nominal scheduled contact phase time, z_0 is the nominal height of locomotion.
- The footstep locations can be used to adjust the speed.
- Posture adjustment on sloped terrain.

Contact Detection

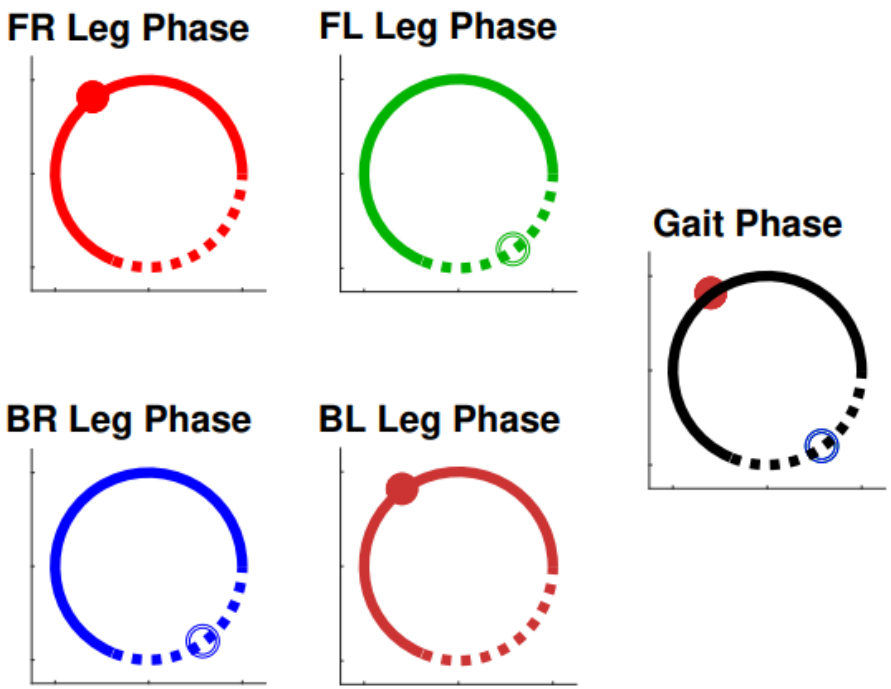


Figure: Gait Scheduler Phase Map.

Contact Detection - State Definition

- Suppose the overall estimated probability of the feet being in contact with the ground is

$$\hat{x}_k = \begin{bmatrix} P_1(c) \\ \vdots \\ P_n(c) \end{bmatrix}.$$

- The probability of contact given the current contact state and subphase percent is chosen to take the form

$$P(c|s_\phi, \phi) = \frac{1}{2}\Big(s_\phi\Big[erf\big(\frac{\phi - \mu_{c_0}}{\sqrt{2}\sigma_{c_0}}\big) + erf\big(\frac{\mu_{c_1} - \phi}{\sqrt{2}\sigma_{c_1}}\big)\Big] + \bar{s}_\phi\Big[2 + erf\big(\frac{\mu_{\bar{c}_0} - \phi}{\sqrt{2}\sigma_{\bar{c}_0}}\big) + erf\big(\frac{\phi - \mu_{\bar{c}_1}}{\sqrt{2}\sigma_{\bar{c}_1}}\big)\Big]\Big),$$

where $s_\phi = 1$ and $s_\phi = 0$ respectively represents scheduled swing state and stance state, ϕ is the scheduled subphase.

Contact Detection - State Equation

$$x_{k+1} = A_k x_k + B_k u_k,$$

where $A_k = 0, B_k = I, u_k = [P_1(c|s_\phi, \phi), \cdots, P_n(c|s_\phi, \phi)]^T$.

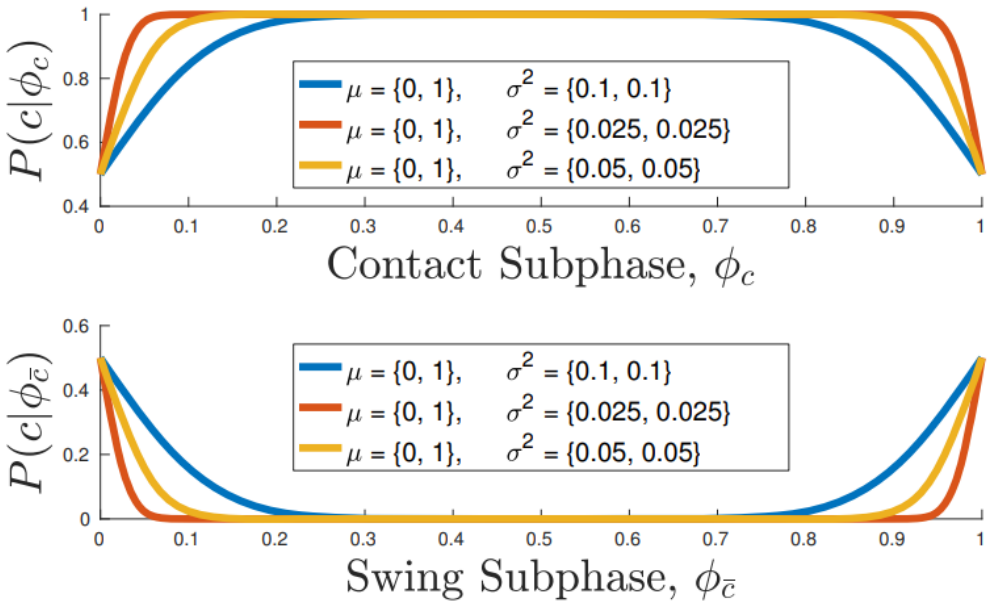


Figure: Phase-Based Probability of Contact.

Contact Detection - Measurement Equation

- Ground height $Z_g \sim \mathcal{N}(\mu_{z_g} = 0, \sigma^2_{z_g})$. The probability of contact given foot height:

$$P(c|p_z) = \frac{1}{2}\left[1 + erf\left(\frac{\mu_{z_g} - p_z}{\sqrt{2}\sigma_{z_g}}\right)\right]$$

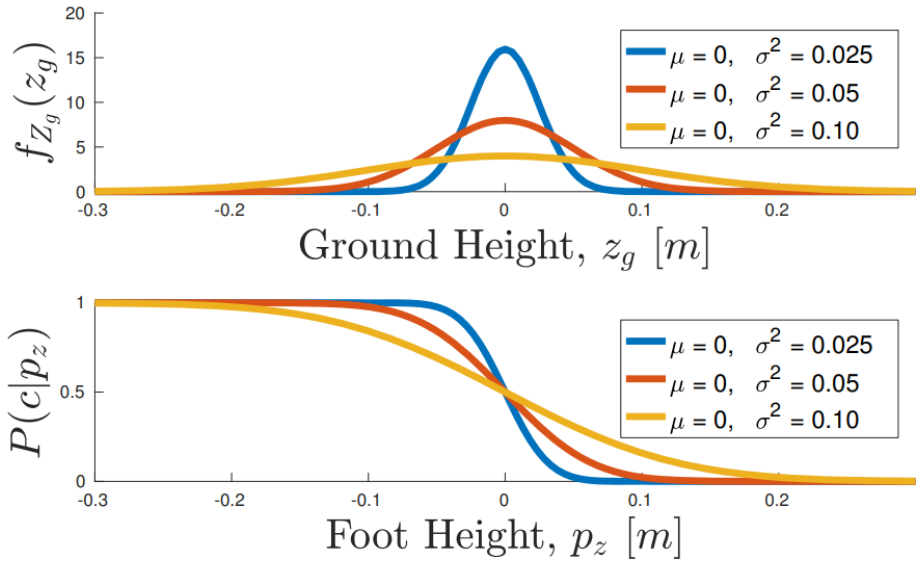


Figure: Probability of Contact From Foot Height.

Contact Detection - Measurement Equation

- Contact force $F_c \sim \mathcal{N}(\mu_{f_c}, \sigma^2_{f_c})$. The probability of contact given estimated foot force:

$$P(c|p_z) = \frac{1}{2}\left[1 + erf\left(\frac{f_z - \mu_{f_c}}{\sqrt{2}\sigma_{f_c}}\right)\right]$$

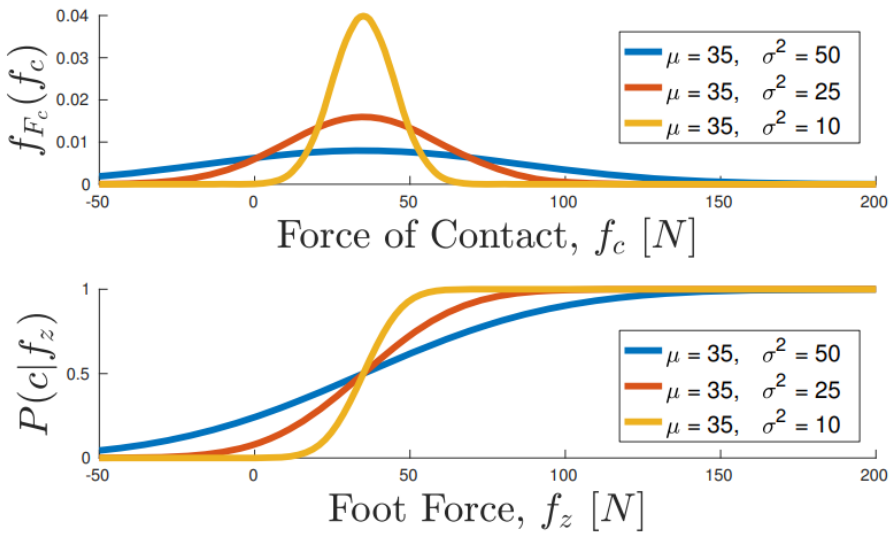


Figure: Probability of Contact From Foot Force.

Contact Detection - Experimental Result

- Use Kalman filter to predict and correction.

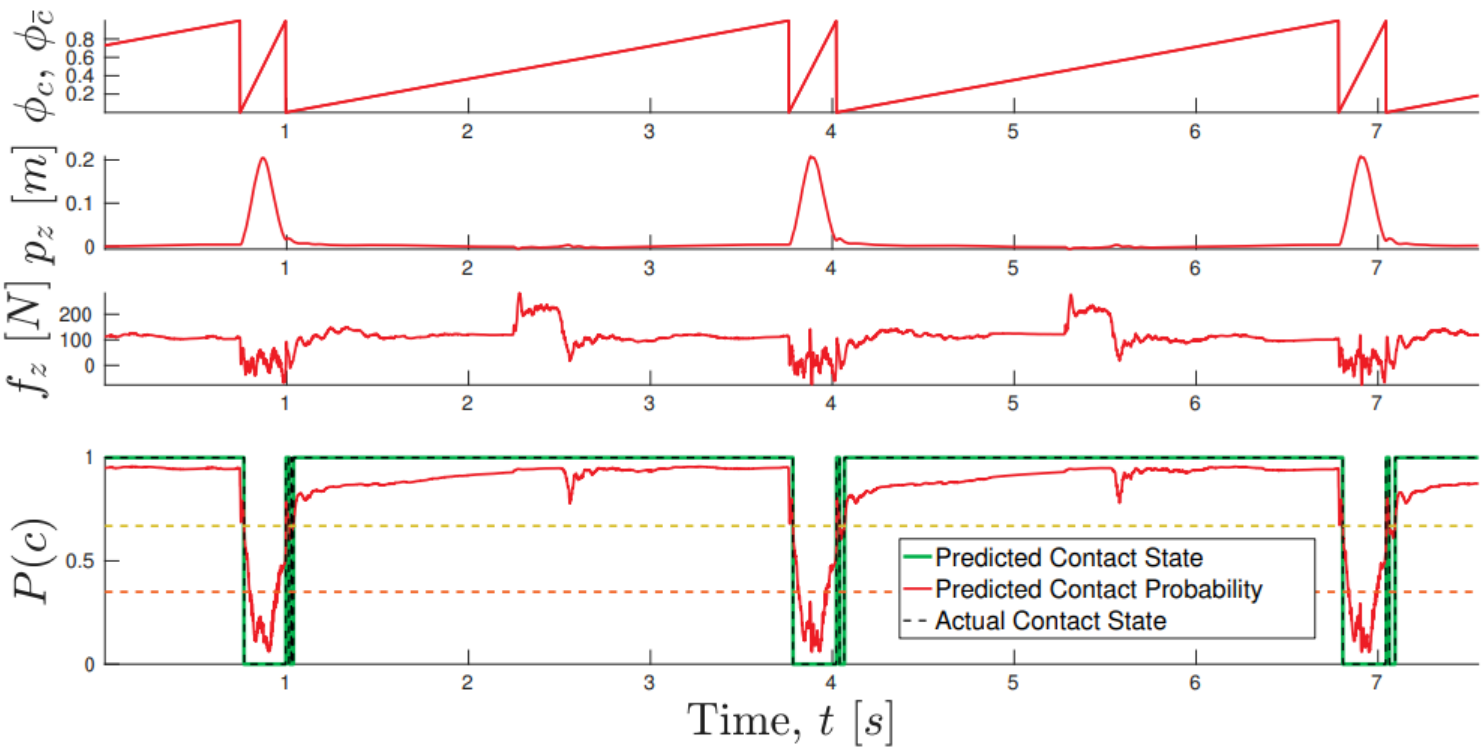


Figure: Experimental Contact Detection.